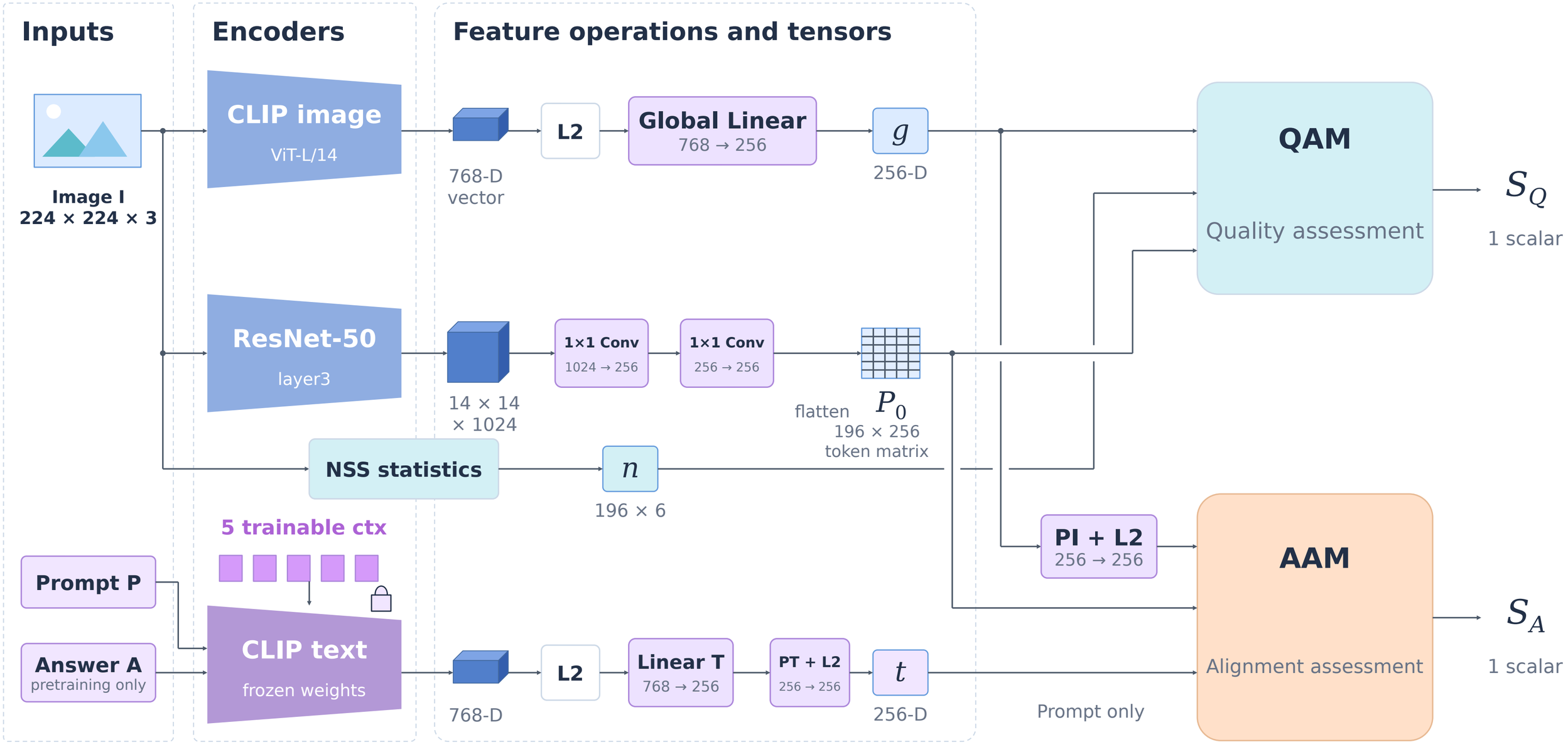
From image tensors to Quality and Alignment scores

projection_only_v1 · dimensions shown for one image · tensor drawings are not to scale



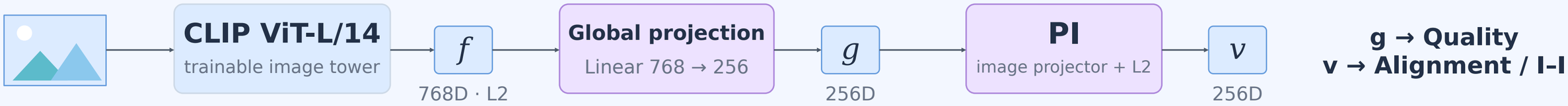
Pretraining only — Two SafeAug views → I–I; Clean view → labels / component anchors / ordinal; Prompt + Answer → SRM relations

Baseline visual tower: trainable. Frozen-CLIP ablation: same shapes and paths. No cross-attention and no fused z. L2 has no learned weights.

SRE | Shared Representation Encoder

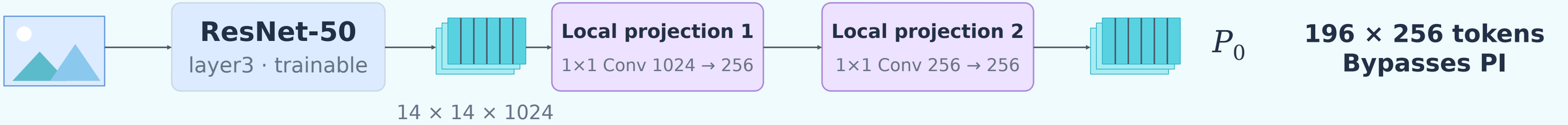
One image produces independent global and local features; text uses five trainable context tokens.

Global visual path

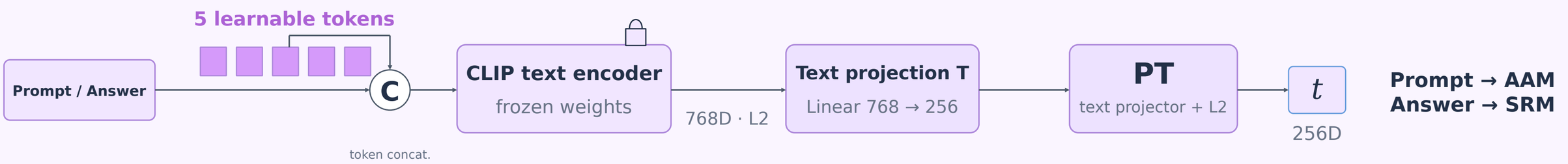


Inside CLIP: 16×16 patches + CLS → 257×1024 tokens → CLS (1024-D) → CLIP output projection (768-D)

Independent local path



Text path



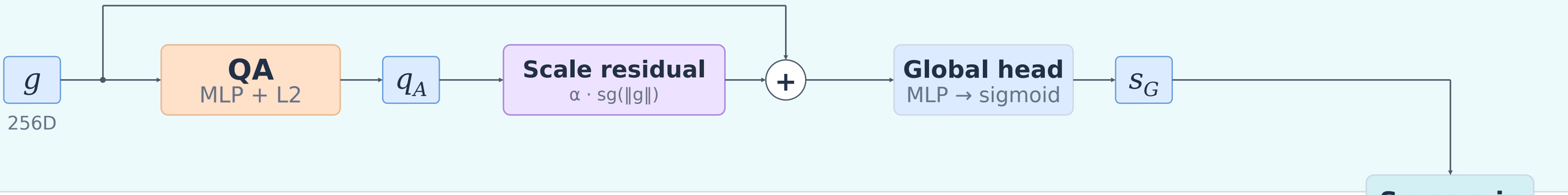


PI/PT: separate 256 → 512 → 512 → 256 MLPs. Shared SRE weights across calls; g goes directly to Quality and P_0 bypasses PI.

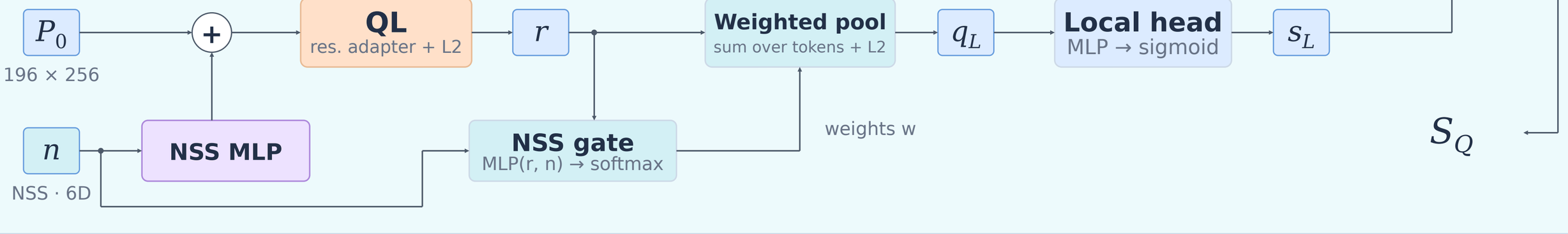
QAM | Quality Assessment Module

Global residual scoring and NSS-guided local scoring remain separate until scalar score mixing.

Global quality



Local quality



Pretraining only · reuse qA and qL above

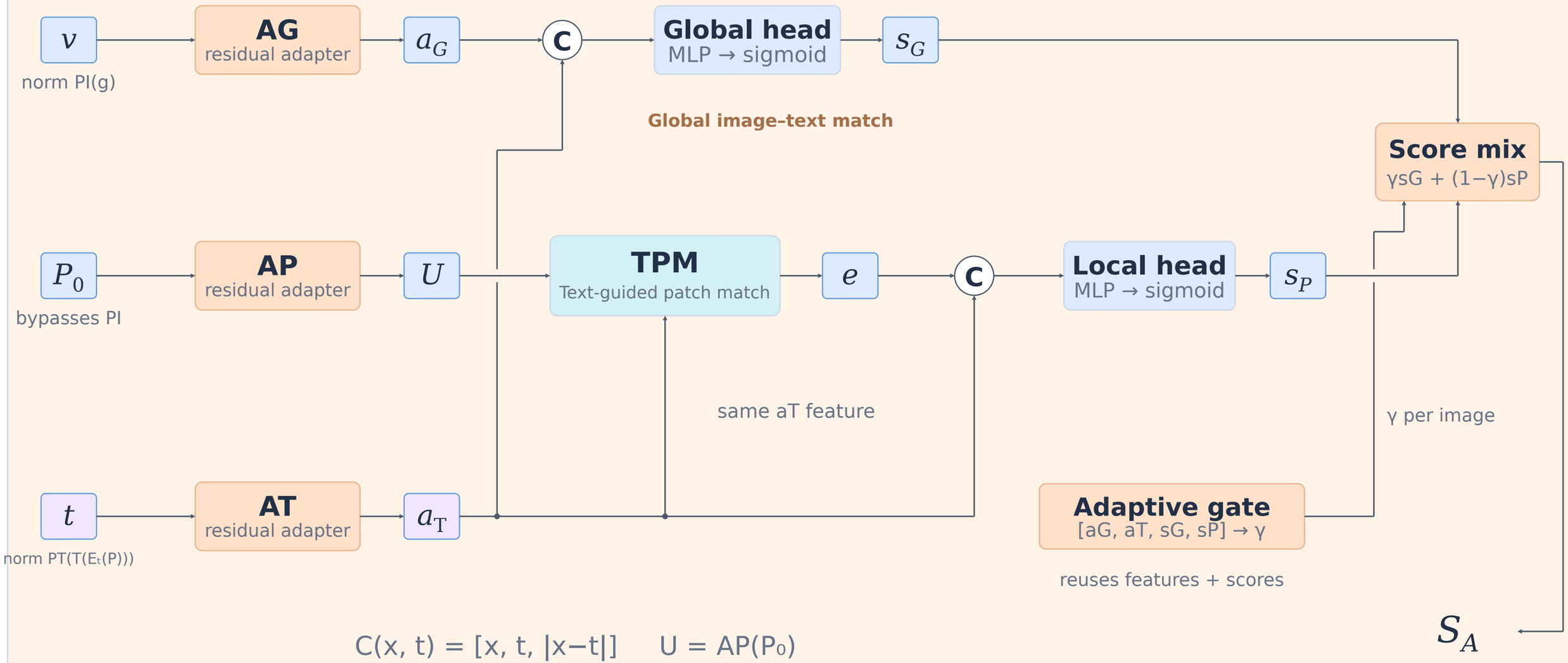


β and α are learned scalars. “+” adds features; score mix combines scalar predictions. Quality text-bank CE stays disabled.

AAM | Alignment Assessment Module

Global matching and text-guided patch evidence produce two scores, combined by an adaptive gate.

Alignment computation

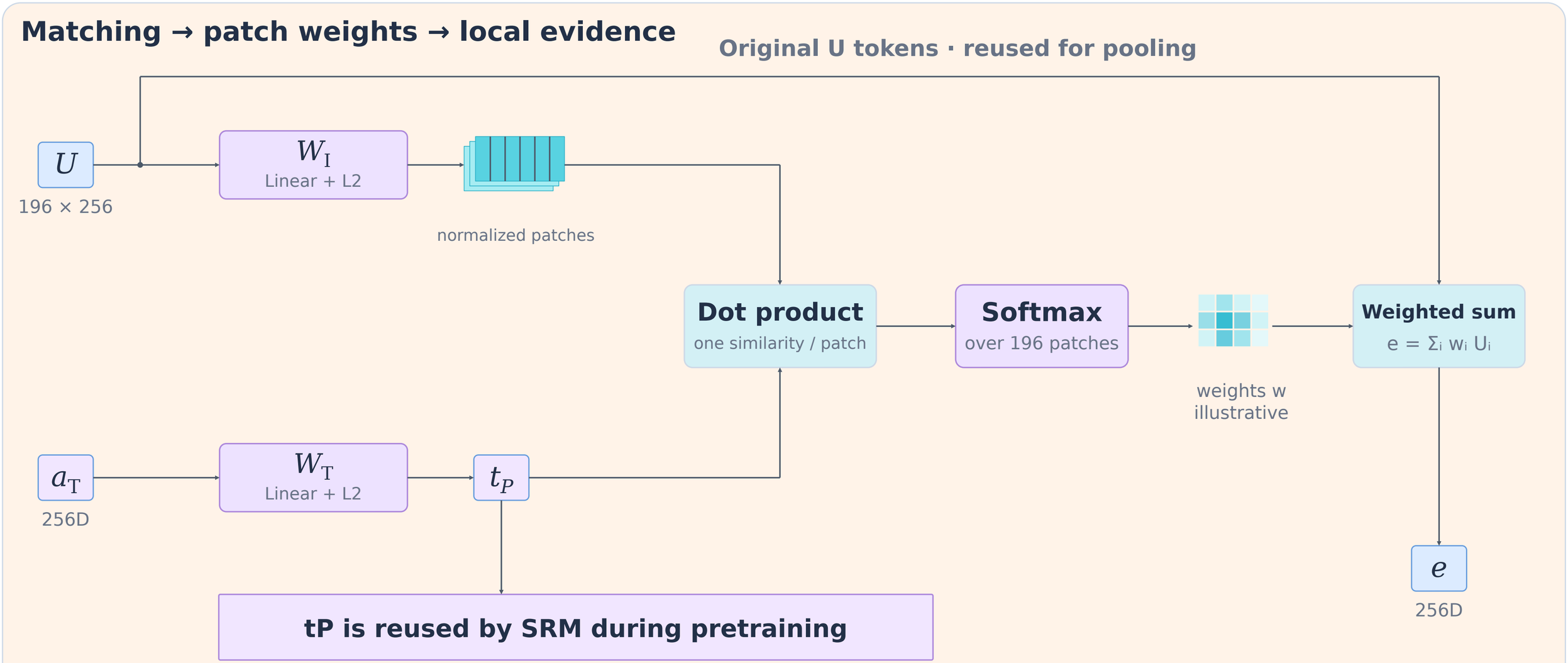


Pretraining taps: aᶜ = norm(aG); tP = norm(W_T(aT)). Answer tA comes directly from SRE, bypassing AAM.

AG / AP / AT are separate adapters. The adaptive gate reuses the four labeled inputs; TPM performs dot-product pooling, not cross-attention.

TPM | Text-guided Patch Matching

Use text to weight patches; pool the original adapted patch tokens to obtain local evidence.

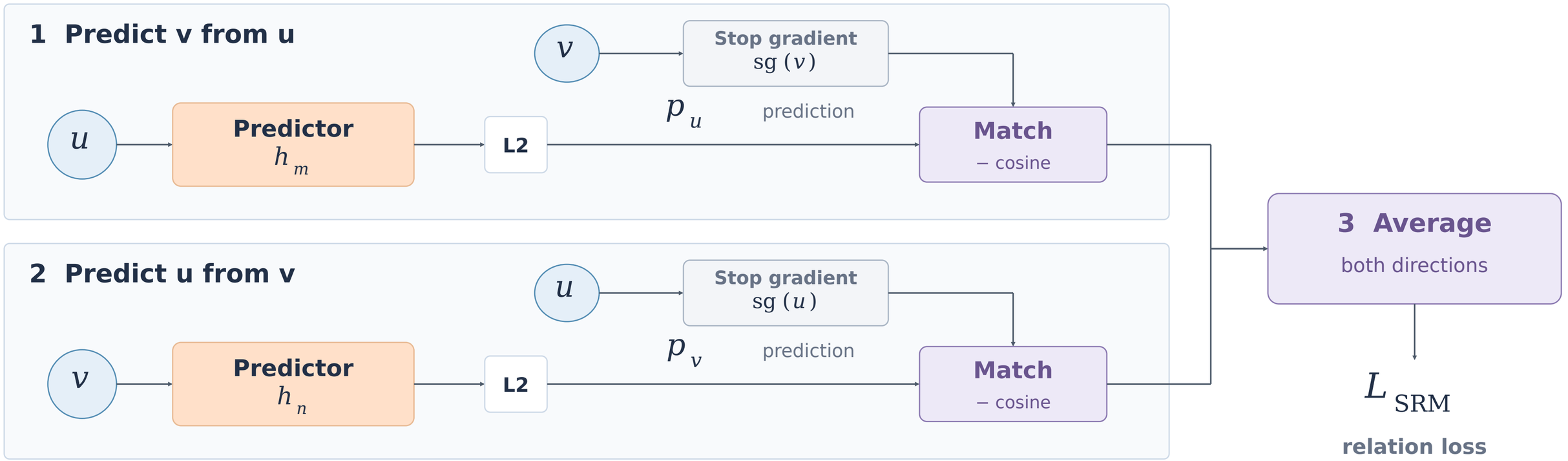


The matching projections produce attention weights only. The weighted values are U = AP(P_0).
This is scalar dot-product patch pooling, with no multi-head cross-attention or global–local fusion output.

W_I and W_T are learned Linear matching layers; both reset for downstream training. Heatmap cells are an illustrative vector schematic.

SRM | Symmetric feature matching

Predict the other feature → swap roles → average the two losses.



Same inputs, opposite roles. Only the prediction path receives gradients.

USE THE SAME MODULE FOR FOUR RELATIONS

Image ↔ Image		Image ↔ Prompt		Image ↔ Answer		Prompt ↔ Answer	
Input pair	Predictors	Input pair	Predictors	Input pair	Predictors	Input pair	Predictors
(v_1, v_2)	$(h_{\mathrm{I}}, h_{\mathrm{I}})$	(a^c, t_P)	$(h_{\mathrm{I}}, h_{\mathrm{T}})$	(a^c, t_A)	$(h_{\mathrm{I}}, h_{\mathrm{T}})$	(t_P, t_A)	$(h_{\mathrm{T}}, h_{\mathrm{T}})$

Only two predictor networks: one for images and one for text, reused across all four relations.

Current mode: positive matching only · 256-D inputs · PI/PT are outside SRM · Targets are detached, not permanently frozen

How to read the tensor operations

L2 is a fixed arithmetic operation; 1×1 projections are learned layers; flatten only changes the layout.

1 L2 normalization · no learned parameters

x = [3, 4] → **x / max(‖x‖₂, ε)** → **[0.6, 0.8]**

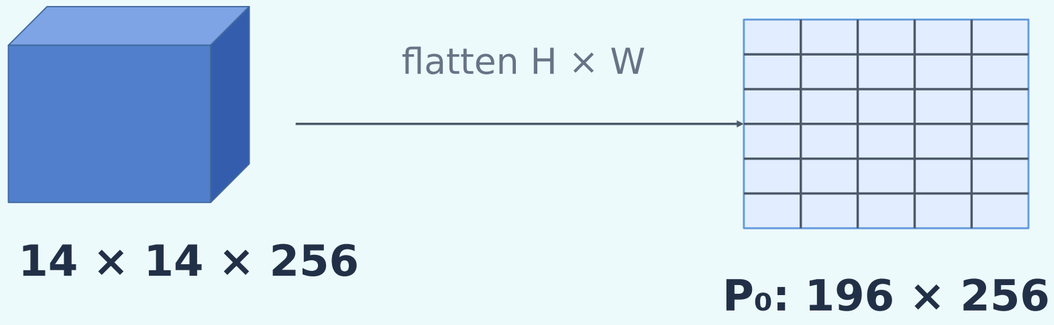
Same shape; divide each feature vector by its L2 norm, with ε protection. No learned weights; not computed across images.

2 Two independent 1×1 convolutions · learned channel projections

14 × 14 × 1024 → **Conv 1: 1024 → 256** → **Conv 2: 256 → 256** → **14 × 14 × 256**

Each location mixes its channel values with a learned matrix. Spatial size stays 14×14; the two layers have different weights.

3 Feature map → token matrix · reshape only



14 × 14 × 256

P₀: 196 × 256

196 rows = 14×14 spatial positions
256 columns = channels / features per position
Total: 196×256 = 50,176 values

A cuboid icon is an illustration of an H×W×C feature map. P₀ itself is a matrix (batch form B×196×256), not a physical cube.